Solar Field Report: Station Meridian-7

Section 1 - Power Generation.

The primary generator at Station Meridian-7 produces 3.5 kilowatts of continuous power under full sunlight. Output is measured at the main bus and logged every fifteen minutes by the telemetry unit. Peak generation occurs between 10:00 and 14:00 local time, when the panel array is closest to perpendicular with the incident sunlight.

Section 2 - Energy Storage.

The backup battery bank stores 12 kilowatt-hours and was installed in March 2021 by the maintenance crew led by Dr. Farah Nazari. The bank uses lithium iron phosphate cells rated for 6000 cycles, chosen over lead-acid for their tolerance of deep discharge in the high daytime temperatures the station regularly experiences.

Section 3 - Water Purification.

Water purification runs on the surplus load and processes up to 400 liters of drinking water per day through a two-stage reverse osmosis membrane. The pre-filter is replaced every ninety days, and the crew keeps three spare membranes in the supply locker.

Section 4 - Logistics.

The nearest resupply depot is 84 kilometers to the north, reachable only during the dry season between May and September. Outside that window the access road floods and deliveries are suspended.

Section 5 - Emergency Protocol.

If generator output drops below 1 kilowatt, switch to battery-only mode and radio the depot on channel 9. Do not attempt repairs on the array during a dust storm; wait until visibility clears.